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An aerial photograph of a white speedboat with a brown cabin, moving across a dark blue ocean. The boat is leaving a white wake behind it. The water's surface is textured with small waves and ripples.

SEALERT

Classifying Maritime Distress Messages with LLM- Generated Data

Review & Refined Project Definition

The challenge is to correctly classify maritime distress messages when ASR transcripts are noisy, incomplete, and lack explicit codewords.

INPUT

Noisy ASR transcript of a
maritime radio communication

Simulated VHF: Distress | Urgency | Safety
Messages



OUTPUT

Classification: One of four severity levels

Distress | Urgency | Safety | Routine

Information Extraction: Structured fields

Location | Vessel Name | Persons on Board | Nature of
Incident | Call Sign / MMSI (optional)

DATA

- 1,872 GPT-4 synthetic maritime messages (468 per class)
- Clean | ASR text (SNR: 18 | 12 | 6 dB)
- With | without codewords
- Audio (TTS) + Whisper ASR
- Multiple styles & scenarios

MODELS

- Transformers: RoBERTa (selected) | DistilBERT
- Baselines: BoW + Logistic Regression | Linear SVM | Naive Bayes
- Infrastructure: GPT-4 | Faster-Whisper | Coqui TTS

METRICS

- F1-score: primary metric | model selection | robustness
- Accuracy: noise-level comparison (Clean | Low | Medium | High)
- Relative change: SNR | WER | codeword masking

Project achievements and novelty

PROJECT ACHIEVEMENTS

- End-to-end system for maritime distress classification from noisy ASR
- Realistic radio-communication pipeline: text → audio → noise → ASR → NLP
- Controlled comparison: Traditional ML vs. Transformer (RoBERTa)

DELIVERABLES

- Severity classification (Distress | Urgency | Safety | Routine)
- Structured operational output (JSON): vessel, location, POB, incident
- Synthetic maritime dataset (text + audio + noisy ASR)

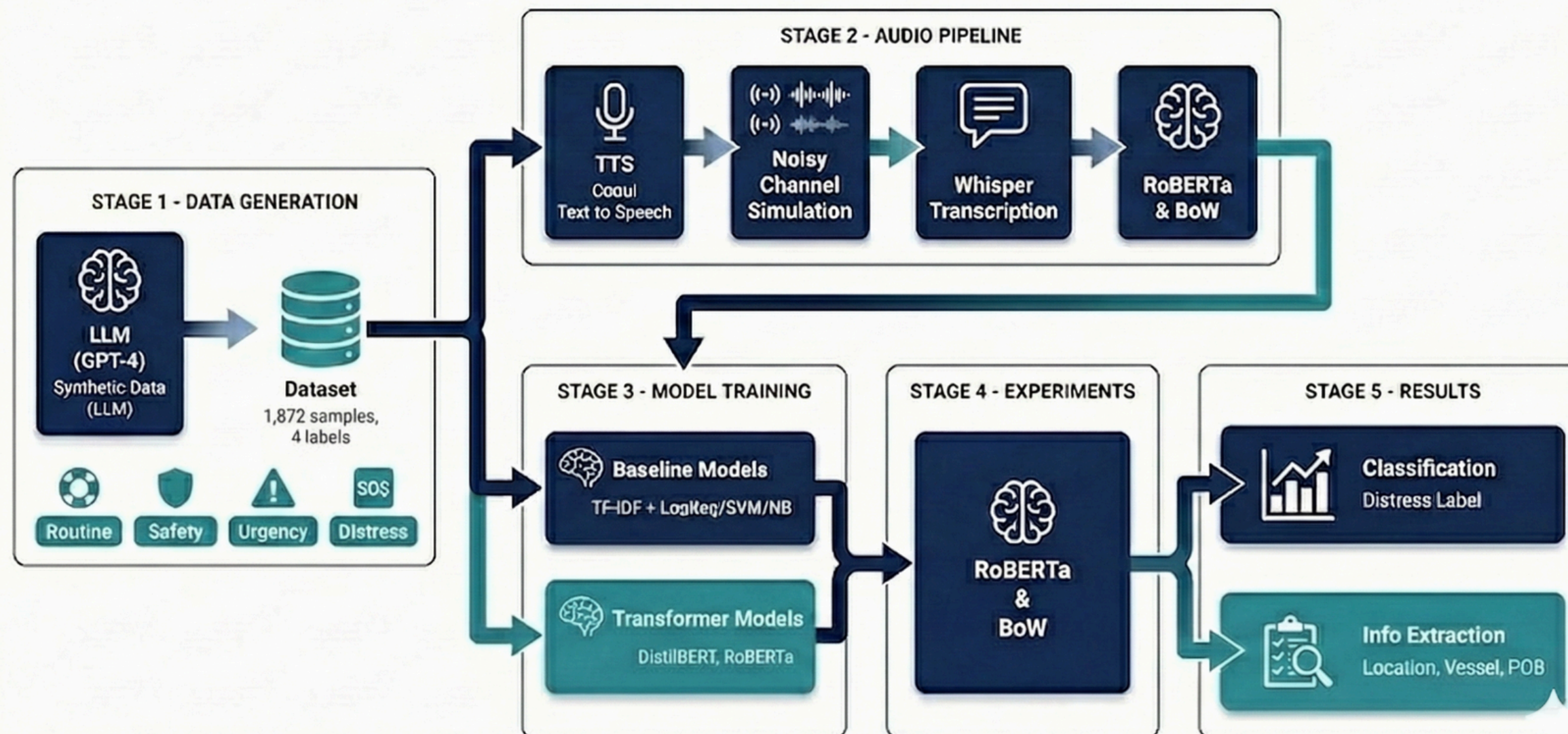


The project's novelty lies in training the system on noisy ASR outputs rather than assuming clean transcripts.

Review Methodology

TEXT → AUDIO → NOISE → ASR → MODEL → STRUCTURED OUTPUT

SeaAlert - Maritime Message Classification



- Synthetic data generation (text + audio)
- Controlled ASR noise simulation (SNR-based)
- Model training & evaluation (F1, Accuracy)
- Robustness & adversarial testing

Example

real example from dataset — demonstrating how severe ASR errors can be under high noise conditions:

ORIGINAL MESSAGE

"MAYDAY, MAYDAY, MAYDAY!
This is the sailing yacht 'Wind
Whisper', uh, we're, um, having a
steering failure—can't control the
rudder. We're drifting, uh,
southeast of, uh, Cape Cod, um,
about 10 nautical miles off the
shore, I think. Weather's, uh, kinda
rough, waves are, like, 3 meters,
and it's, um, getting windy. We've
got, uh, three people onboard.
Need assistance, like, right now!
Over."



ASR OUTPUT HIGH NOISE

"maybe, maybe, maybe, this is
the sailing yacht with this boat
out here, I'm having a steering
failure, can't control the rudder,
we drifting, I've surfaced through,
I kick hard, I'm about 10 knots, I
come out as off as sure I think,
where those are, can't laugh,
waves are like three meters, and
it's, I'm getting in the, we've got a
few people on board, need the
systems right right now, over."

MODEL OUTPUT

Classification



DISTRESS

Extracted Information

- Vessel: Wind Whisper
- Location: SE of Cape Cod,
~10 nm offshore
- POB: 3
- Nature: Steering failure,
drifting

Critical ASR Errors Shown:

MAYDAY, MAYDAY, MAYDAY → maybe, maybe, maybe ● (codeword completely lost!)

Wind Whisper → with this boat out here (vessel name destroyed)

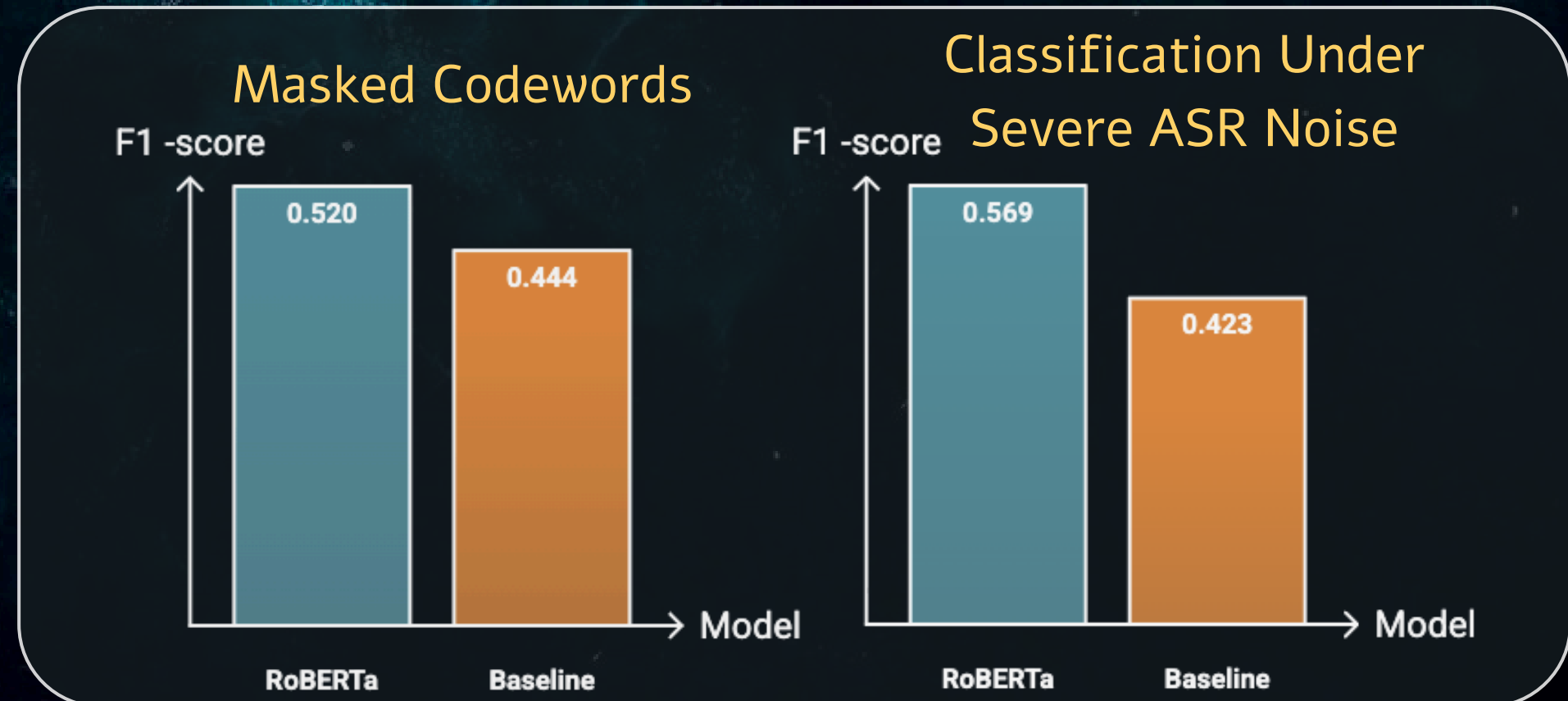
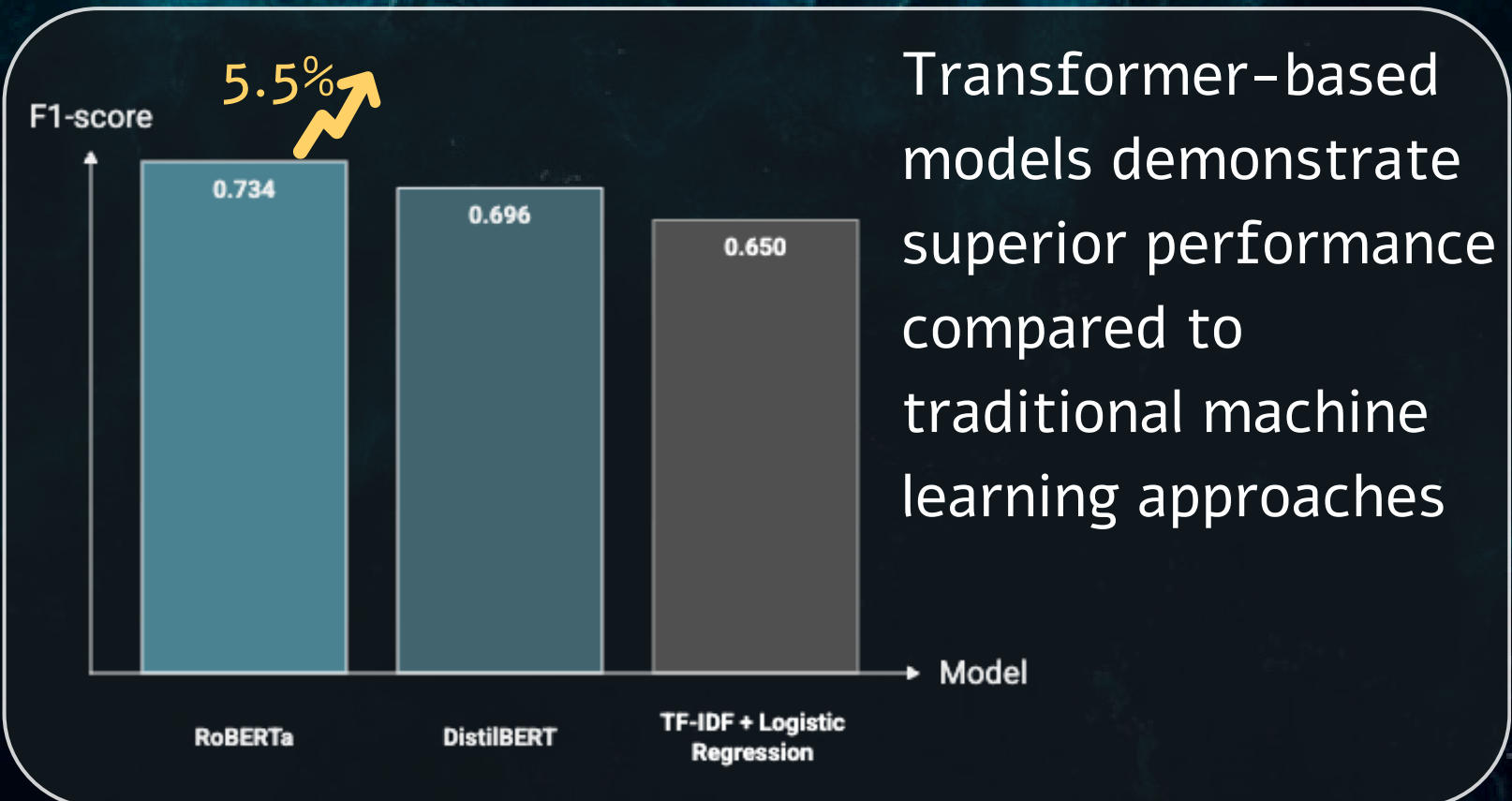
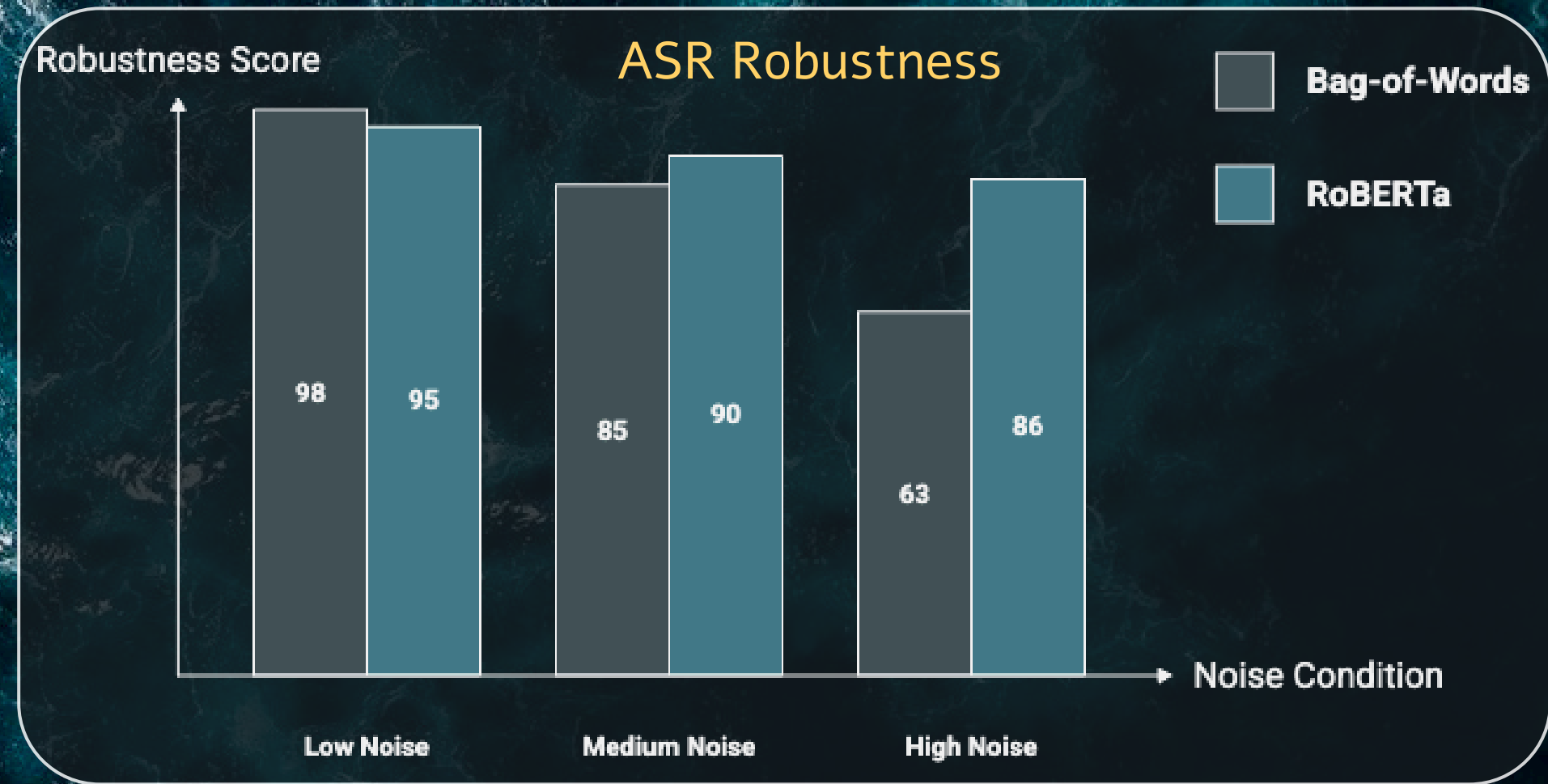
Cape Cod → kick hard (location corrupted)

nautical miles off the shore → knots, I come out as off as sure (nonsensical)

Results

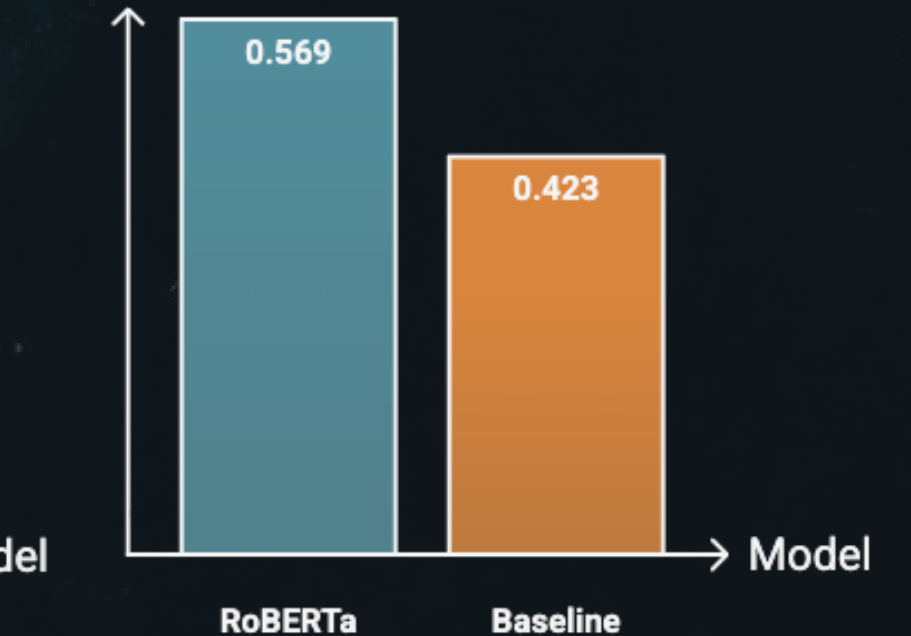
- Best F1-score achieved: 0.734 (RoBERTa)
- Transformers outperform traditional models
- Performance remains stable under noisy ASR conditions

RoBERTa delivers the best overall performance and robustness for noisy maritime ASR data.



Classification Under Severe ASR Noise

F1 -score



Conclusion & Future Work

- ✓ The system successfully implements an end-to-end pipeline that transforms noisy ASR transcripts into a structured JSON output, including vessel name, location, and nature of distress
- ✓ Demonstrated improvement: Enhanced contextual understanding and effective handling of adversarial traps
- ✓ Reliable detection of maritime distress from noisy ASR text
- Remaining challenge: Continued reliance on explicit codewords (e.g., MAYDAY)

IMPROVEMENT ACTIONS

- Increase Data Augmentation with Noisy ASR
- Reduce Dependence on Codewords
- Strengthen Contextual Learning
- Mitigate Error Propagation Across the Pipeline
- Re-evaluate Models Under Realistic Noise Conditions

ADDITIONAL EXPERIMENTS

- Fine-tune ASR for Maritime Vocabulary
- Add Adversarial Data Augmentation
- Implement Multi-Task Learning (Classification + NER)
- Replace Rule-Based Extraction with Trained NER
- Scale and Diversify the Dataset
- Validate on Real Maritime Radio Data
- Extend to Additional Languages and Acoustic Features